

Carbon border mechanism → which is a unit's carbon.
→ visa → → already.

CBAM Platform — Team Handbook

Part 1: What CBAM is, from zero · Part 2: What we build, in what order, by when

Vlookup Business Solutions · For the development team (no prior knowledge assumed) · 29 September 2026

How to use this handbook

Part 1 (Sections 1–6) teaches you CBAM from nothing — read it fully before touching Part 2. Part 2 (Sections 7–12) is the development plan: releases, week-by-week schedule, team, and standing rules. Deep regulatory reference lives in the companion document CBAM-Complete-Workflow.docx; architecture and data model live in CSorted-Platform-Technical-Specification.docx (Parts A3, C3, D, E).

PART 1 — CBAM EXPLAINED FROM ZERO

CBAM in one sentence:

From 1 January 2027, UK businesses that import certain carbon-heavy goods (steel, aluminium, cement, fertilisers, hydrogen) must pay a tax based on how much carbon dioxide was released when those goods were made in the overseas factory — and it is the IMPORTER's job to find out that number, prove it, calculate the tax, and pay HMRC (the UK tax authority).

1 — Why does this tax exist? (the story)

Making steel, cement or aluminium releases a lot of carbon dioxide (CO₂). The UK already charges its own factories for their CO₂ through a scheme called the UK ETS (Emissions Trading Scheme) — a UK steel plant pays for every tonne of CO₂ it emits.

Problem: a factory in another country with no carbon charge can make the same steel cheaper, and UK buyers would simply import from there instead. UK factories lose business, and the world's CO₂ doesn't go down — it just moves country. This is called 'carbon leakage'.

The fix: CBAM — the Carbon Border Adjustment Mechanism. It puts an equivalent carbon charge on IMPORTED goods at the border, so imported steel and UK-made steel carry a similar carbon cost. Fair competition, and a reason for overseas factories to clean up too.

Analogy for the team:

Think of it like a visa fee based on behaviour. UK factories already 'pay for their pollution' at home. CBAM makes imported goods pay an entry fee sized to the pollution created making them. Clean factory = small fee. Dirty factory (or factory that refuses to say) = big fee.

2 — Who is caught by it?

Question	Answer
Which goods?	Only 5 sectors: <u>aluminium, cement, fertilisers, hydrogen, iron & steel</u> . Each good is identified by its customs 'CN code' (an 8-digit product code on every import). If the CN code is on HMRC's CBAM list, the good is in scope
Which businesses?	Any UK importer whose <u>CBAM goods are worth £50,000 or more</u> — tested two ways at the same time: (a) looking back over any <u>rolling 12 months</u> , and (b) looking forward: expected to cross £50k in the next 30 days. Cross either test → must register with HMRC within 30 days
Who is liable?	The <u>importer</u> (called the 'liable person'). Not the overseas factory, not the shipping company. And a named human at the importer — the <u>Senior Accounting Officer (SAO)</u> — signs the returns and is personally responsible for their accuracy
When?	Tax applies to goods imported from <u>1 January 2027</u> . First return covers the whole of 2027 and is due <u>31 May 2028</u> . After that: quarterly returns

3 — The key idea: the tax depends on a number only the overseas factory knows

The tax is calculated from 'embedded emissions' — how many tonnes of CO₂ were released to make each tonne of the product. That number depends on the specific factory: a modern electric-arc steel plant might emit 0.4 tonnes of CO₂ per tonne of steel; an old coal-fired one might emit 2.0+.

The importer has two options:

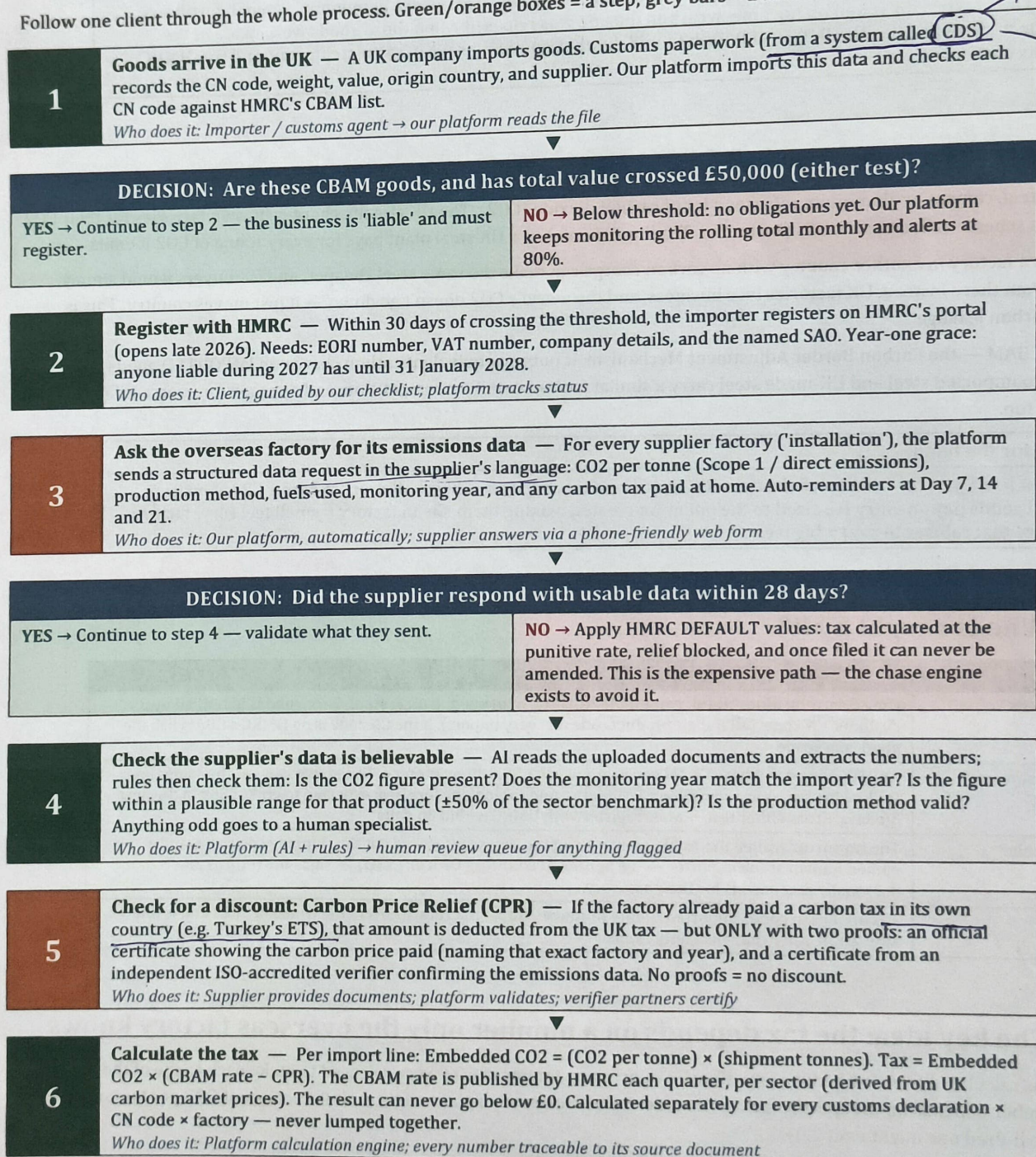
Option	What it means	Consequence
Option A — Actual data	Ask the overseas factory for its real, measured emissions number (with evidence)	Usually a MUCH lower tax. Also unlocks 'Carbon Price Relief' — a discount for any carbon tax the factory already paid in its own country
Option B — Default values	Use HMRC's published standard number for that product type	Deliberately set HIGH (worst-case). No relief allowed. And once a return is filed using defaults, it can NEVER be corrected later — the high tax is locked in forever

This is the single most important fact about CBAM — and about our product:

Real worked example: 48.2 tonnes of steel from a modern Turkish plant. With actual verified data the tax is £785.25. With HMRC defaults it is £4,771.80 — six times more, on ONE shipment. Multiply across a year of shipments and getting real supplier data is worth tens of thousands of pounds per client. Our platform exists to win this data race. Everything else is admin around it.

4 — How CBAM works, start to finish (the flowchart)

Follow one client through the whole process. Green/orange boxes = a step; grey bars = a decision with two paths.



importer API
CDS have the data of all the importer detailed way.

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Prepare and approve the return — The platform assembles the return: one line per declaration/CN/factory, showing weight, emissions, rate, relief and tax, with every line linked to its evidence. The client's SAO reviews and personally approves — a human click, always, because the SAO carries personal legal liability.

Who does it: Platform drafts; ops team checks; client SAO approves

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File with HMRC and pay — Return filed, tax paid. First period: all of 2027, due 31 May 2028; then quarterly. All evidence kept for 6 years — HMRC can audit at any time, and the platform can produce the full evidence pack for any line in one click.

Who does it: Client files (platform produces exactly what to enter); platform archives everything

DONE — then the cycle repeats for every new shipment and every new quarter. The platform never stops monitoring the threshold, chasing suppliers, and watching deadlines.

5 — Jargon translator (keep this open while reading anything CBAM)

Term	Plain English
CBAM	Carbon Border Adjustment Mechanism — the import carbon tax itself
HMRC	His Majesty's Revenue & Customs — the UK tax authority we ultimately file to
CN code	8-digit customs product code on every import — tells us if the good is in CBAM scope
CDS	Customs Declaration Service — HMRC's system holding every import record; our main data source
EORI	A business's customs ID number — needed to register
Installation	The specific overseas FACTORY that made the goods (not the trading company that sold them). Tax is calculated per factory
Embedded emissions	Tonnes of CO2 released making each tonne of product — the number the whole tax hangs on
Scope 1	Direct emissions from the factory itself (fuel burned on site). The only emissions taxed from 2027; electricity-related ('Scope 2') joins from 2029
Default values	HMRC's deliberately-high standard emission numbers, used when real data is missing. The expensive path
CBAM rate	The £-per-tonne-of-CO2 price, set by HMRC per sector per quarter
CPR — Carbon Price Relief	Discount for carbon tax already paid in the factory's home country. Needs certificates + independent verification
Verifier	Independent inspection company (e.g. Bureau Veritas, SGS) accredited to ISO 17029 + ISO 14065, who certifies a factory's emissions numbers
SAO	Senior Accounting Officer — the named person at the client who signs returns and is personally liable
Liable person	Legal term for the importer who owes the tax

6 — So what exactly does our tool automate?

Painful manual reality (without us)	What our platform does instead
Someone re-types customs paperwork into spreadsheets, shipment by shipment	Imports the CDS file; every line classified against the CBAM list automatically
Nobody notices the £50k threshold until HMRC writes a letter	Rolling total recalculated nightly, both tests; alert at 80%, alarm at breach with the 30-day clock
Emailing 30 factories in 6 countries, in English, and forgetting to follow up	Structured multilingual requests, phone-friendly supplier form, automatic Day 7/14/21 chasing, Day 28 escalation
Squinting at a Turkish PDF monitoring report trying to find the CO2 number	AI extracts the numbers into structured data; rules check plausibility; humans review only the flagged ones
A spreadsheet formula someone built once and nobody dares touch	A tested calculation engine where the worked example above is a permanent automated test
A shoebox of PDFs when HMRC audits	Every figure on every return linked to its source document; 6-year evidence pack in one click

PART 2 — DEVELOPMENT KICKOFF PLAN (13 WEEKS TO GO-LIVE)

THE ONE THING TO UNDERSTAND BEFORE ANY CODE IS WRITTEN

UK CBAM goes live 1 January 2027 — 13 weeks from today. But the first tax return is not due until 31 May 2028. So we do NOT need the tax calculator in January. What we DO need live on 1 January is the DATA CAPTURE MACHINE: imports logged, the £50,000 threshold monitored, and supplier emissions-data outreach firing. 2027 emissions data cannot be collected retroactively — a client who misses the collection window pays roughly 6× more tax. The data race is the product. The calculator comes second.

7 — Regulatory status check (Sept 2026) — what the platform must reflect

- The rulebook is now essentially final: three statutory instruments laid 13 July 2026 (Administrative Provisions; Rate Calculation & CPR — SI 2026/809; Transitory Provisions), in force 1 Jan 2027. HMRC guidance published August 2026.
- HMRC registration opens via Government Gateway in Q4 2026 — weeks away. Importers liable before it opens must keep full records from day one.
- HMRC has issued a Carbon Price Verification Form for CPR claims (installation details, scheme details, emissions data). Our supplier portal fields must map onto this form — get the form, mirror it.
- Rate = average UK ETS auction price for the preceding quarter minus a free-allowance adjustment, set per sector, published quarterly. Store rates as reference data per sector per quarter, never hard-code.
- Still draft: the Emissions & Verification Regulations (consulted Apr–May 2026). Build validation rules as configurable data, not code, so we absorb the final version without a release.

8 — Build order: three releases, dated by law not by convenience

Release	Deadline	What ships	Why this date
R1 — Data Capture <i>phase 2</i>	1 Dec 2026 (9 weeks)	Imports register (CSV import of CDS data + manual entry) · CN-code scope checker · rolling £50k threshold monitor (both tests: backward 12-month AND forward 30-day) · client registration checklist (EORI, VAT, SAO) · supplier + installation register · supplier outreach engine with Day 7/14/21/28 chase · tokenised supplier portal (form mirrors the EU template + HMRC Carbon Price Verification Form) · document upload + storage · ops dashboard · audit log	Clients onboard in December, register on the HMRC portal when it opens, and outreach fires to every supplier BEFORE 1 Jan. This is the release that wins or loses the client's money
R2 — Validation & Intelligence	28 Feb 2027	Claude API document extraction (monitoring reports, ETS certificates, verifier certs → structured JSON) · validation rules engine (~10 rules from Part 1 step 4: Scope 1 present, period matches, ±50% plausibility, method valid, cert names installation, verifier ISO 17029+14065, precursor consistency) · human review queue · RAG compliance dashboard per client · CPR evidence tracker	Supplier responses arrive Jan–Mar in volume. Validation must keep pace so bad data is challenged while suppliers are still engaged
R3 — Calculation & Filing	30 Sep 2027	Calculation engine (pure function: embedded × (rate – CPR), floor £0, per declaration × CN × installation) · quarterly rate reference admin · liability forecasts per client · return generator with SAO approval workflow · evidence pack generator · the hard locks (defaults filed = permanent, defaults block CPR)	First return due 31 May 2028. Shipping R3 by Sept 2027 gives a full quarter of dry-runs with real client data before anything is filed

Rule for the team: nothing from R3 gets built during R1. No matter how interesting the calculator is.

The temptation will be to build the tax engine first because it is the intellectually satisfying part. Resist it. An empty calculator in January is worthless; a full imports register with outreach running is the entire business.

— Week-by-week plan to R1 (the next 9 weeks)

Weeks	Workstream	Deliverable / gate
W1 (29 Sep – 3 Oct)	Setup + reading	Team reads this handbook + companion docs. Repos created (app, api), CI/CD, Supabase project (EU region), envs (local/staging/prod), Sentry. GATE: everyone can explain the £50k two-test threshold and why defaults are poison
W2–3	Foundations	Schema per tech-spec Part C (shared core + cbam_ tables), Row Level Security policies + tests, auth (ops/client roles), INSERT-only audit_log, seed data: CBAM CN codes list, sectors. GATE: RLS proven — client A cannot read client B in an automated test
W3–4	Imports register	CSV import of CDS export format (get a real sample file from a freight forwarder — Jenny's task), field validation, CN scope check against seeded list, manual entry form, imports list UI. GATE: 500-row CDS file imports cleanly with errors reported per row
W4–5	Threshold monitor + client onboarding	Nightly rolling-12-month value calc per client + 30-day forward test, alert at 80% and at breach, registration checklist UI (EORI/VAT/SAO capture). GATE: threshold maths verified against 5 hand-calculated scenarios
W5–7	Supplier outreach + portal	Supplier/Installation register, data_request creation, multilingual email templates (EN/TR/ZH/HI/DE first), Resend integration, chase scheduler (Celery), tokenised portal: structured form + file upload, mobile-first. GATE: full Day 7/14/21/28 sequence runs in staging (minutes-mode) end to end
W7–8	Ops dashboard + documents	Per-client view: imports, threshold status, outreach status per supplier, documents received, deadline list. Document storage with versioning. GATE: ops can answer 'where is client X?' in one screen
W8–9	Hardening + pilot onboarding	Pen-test basics, backup/restore drill, load test (10k imports), UAT with Jenny + ops team using the worked example as the test script. GATE: R1 sign-off — a real pilot client's data loaded and outreach sent

10 — Team shape and who owns what

Role	Count	Owns	First week task
Tech lead (senior full-stack)	1	Architecture, RLS/security, code review of anything touching money or legal state, release gates	Repo + Supabase setup; schema review against tech spec Part C
Backend dev (Python)	1–2	FastAPI, Celery chase engine, threshold calc, CSV import pipeline	CDS import parser spike with a real sample file
Frontend dev	1	Ops dashboard, client portal, supplier portal (mobile-first)	Auth flows + app shell
QA (can be part-time)	1	Threshold-maths test suite, RLS tests, chase-sequence simulation, UAT scripts	Write the 5 hand-calculated threshold scenarios
Domain owner — Jenny + ops	—	CN code list accuracy, email template wording, HMRC form mapping, UAT, supplier language review	Obtain: real CDS export sample, HMRC Carbon Price Verification Form, EU Communication Template

Minimum viable team: 3 devs + QA + Jenny. If you have fewer, R1 scope holds and R2 slips — never the reverse.

11 — Standing rules (print these on the wall)

- The law is data, not code. Rates, defaults, CN codes, validation thresholds, chase schedules — all reference tables editable by ops, never constants in code. The Emissions & Verification Regulations are still draft; we absorb changes without releases.
- Legal state transitions are one-way. A filed return locks. Defaults applied after Day 28 lock. Build these as database constraints AND application checks — not UI disablement.
- Every state change writes to the INSERT-only audit log with actor, before/after, timestamp. Six-year retention. This is a feature we sell, not plumbing.
- The worked example (48.2t Turkish EAF steel: £785.25 actual vs £4,771.80 default) is the permanent golden test. It runs in CI forever. Any calculation change that breaks it blocks the merge.
- Suppliers get magic links, never passwords. If a factory manager in Turkey can't complete the form on a phone in under 10 minutes, the portal has failed — test exactly that.
- Nothing auto-files, ever. AI extracts, rules validate, humans approve. The SAO's personal liability means a named human clicks every filing.
- Weekly demo every Friday against the gates in Section 9 — not against task lists. A gate not met = we talk scope, not extend dates. 1 January does not move.
- British English in all user-facing text. Dates '14 March 2027'. Currency £ with thousands separators.

12 — What happens in parallel (not dev work, but blocking)

When	Action	Why it blocks dev
This week	Obtain: a real CDS export file (from a freight forwarder contact), the HMRC Carbon Price Verification Form, the EU Commission Communication Template (Excel)	W3 import parser and W5 portal form fields are designed from these real artefacts, not guesses
This week	Decide product/brand name for the CBAM tool — needed for email domain, portal URL, templates	W5 outreach emails carry the brand; changing it later means re-sending to suppliers
October	Chamber briefing 'UK CBAM: 90 days to go-live' + recruit 2-3 pilot clients (free/discounted for feedback + case study)	W8-9 UAT needs a real client's real import data; and pilots must register when the HMRC portal opens in Q4
October	Contact 2 verifier bodies (Bureau Veritas, SGS) re referral partnership	R2 CPR validation needs to know what real verifier certificates look like
November	Pilot clients onboarded on R1; their supplier lists loaded; outreach queued to fire 2 Jan	The entire January value proposition

Companion documents: *CBAM-Complete-Workflow.docx* (regulatory detail) · *CSorted-Platform-Technical-Specification.docx* (architecture & data model). Questions logged against section numbers. Weekly demo: Fridays, against Section 9 gates.